INTERIM PROGRESS REPORT - 2

on

A ROBUST DEEP NEURAL NETWORK MODEL FOR AUTOMATED FUNDUS IMAGE ANALYSIS AND DIABETIC RETINOPATHY DETECTION

Submitted by

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ABSTRACT

Diabetic Retinopathy (DR) is a serious eye disease caused by long-term diabetes that damages the blood vessels in the retina and can lead to permanent vision loss if not detected early. Early diagnosis is essential to prevent severe complications, but traditional screening methods require skilled ophthalmologists and specialized equipment. This makes large-scale screening difficult, especially in rural areas. Therefore, automated systems using artificial intelligence and deep learning have become important tools for assisting medical professionals in detecting diabetic retinopathy from retinal fundus images.

In this project, a hybrid deep learning framework is proposed for automated diabetic retinopathy detection and classification. The system combines EfficientNet-B3 and ConvNeXt-Tiny architectures to extract deep retinal features and integrates clinical image features such as intensity statistics, vessel strength, and entropy. These features are fused to create a comprehensive representation of retinal abnormalities. The model is trained on the APTOS 2019 dataset and achieves an accuracy of 85.82%, demonstrating improved performance and reliability for automated DR screening.

Sustainable Development Goals (SDGs) Supported:

1) SDG 3 – Good Health and Well-Being**:** Your system enables early detection of diabetic retinopathy, helping prevent blindness and improving quality of life through accessible AI-assisted screening.
2) SDG 9 – Industry, Innovation and Infrastructure: The project applies advanced AI, deep learning, and medical image processing, promoting innovation in healthcare technology and smart diagnostic systems.
3) SDG 10 – Reduced Inequalities: Automated DR screening can be deployed in rural and low-resource areas where ophthalmologists are scarce, reducing healthcare inequality.

Fig 1. Sustainable Development Goals of 3,9,10

INTRODUCTION

Diabetic Retinopathy (DR) is a progressive eye disease caused by prolonged high blood sugar levels that damage the small blood vessels in the retina. The retina is a light-sensitive tissue located at the back of the eye that plays an essential role in vision. When these blood vessels are damaged, they may leak fluid or blood, causing swelling and abnormal vessel growth. If the disease is not detected and treated in its early stages, it can lead to severe vision impairment or permanent blindness. Therefore, early diagnosis of diabetic retinopathy is extremely important.

The increasing number of diabetes patients worldwide has significantly raised the risk of diabetic retinopathy cases. Early screening and continuous monitoring are necessary to prevent complications. However, traditional diagnosis requires ophthalmologists to manually examine retinal fundus images using specialized medical equipment. This process is time-consuming and requires highly trained professionals. In many rural or low-resource areas, access to advanced diagnostic facilities and eye specialists is limited. As a result, many patients remain undiagnosed until the disease reaches an advanced stage, making treatment more difficult and less effective.

With recent advancements in artificial intelligence, deep learning techniques have become powerful tools for automated medical image analysis. Convolutional Neural Networks (CNNs) can automatically learn complex patterns from retinal images and identify disease-related features such as microaneurysms, hemorrhages, and lesions. In this project, a hybrid deep learning framework is proposed that integrates EfficientNet-B3 and ConvNeXt architectures along with clinical feature extraction. The model is trained on the APTOS 2019 dataset and aims to improve the accuracy and reliability of automated diabetic retinopathy detection systems.

LITERATURE SURVEY

Several studies have explored automated diabetic retinopathy (DR) detection using machine learning and deep learning techniques. Yi et al. [1] proposed a DR diagnosis method based on the RA-EfficientNet architecture, which uses residual attention mechanisms to improve lesion detection in retinal images. Their model achieved improved classification accuracy, but it relied on a single deep learning architecture and did not incorporate multiple feature extraction techniques to enhance the robustness of the system.

Balaji et al. [2] introduced a hybrid DR classification framework that combines CNN, ResNet, and GoogLeNet models for retinal image analysis. Their approach improved classification performance by utilizing multiple convolutional neural networks. Similarly, Kobat et al. [3] developed a deep learning-based automated DR detection system that achieved promising results. However, both approaches mainly relied on deep features and did not include clinical image features that could improve interpretability and diagnostic reliability.

Manasa et al. [4] proposed a machine learning-based DR detection system using retinal images and conventional classification algorithms. Although the method demonstrated moderate performance, traditional machine learning models often fail to capture complex hierarchical patterns present in medical images. Rautaray et al. [5] later introduced a FastViT-based knowledge distillation approach for DR detection, achieving approximately 85.5% accuracy, but lightweight architectures may reduce deep feature representation.

Karthik et al. [6] developed a deep learning framework for early detection and severity classification of DR, achieving about 79% accuracy. Several review studies such as Vij and Arora [7], Agarwal and Bhat [8], and Bhoyar and Patel [9][10] highlighted the growing role of deep learning in DR detection. These studies emphasize the importance of hybrid architectures and feature fusion strategies, which can improve accuracy and provide more reliable automated retinal disease screening systems.

SYSTEM MODEL DESCRIPTION

The proposed methodology focuses on developing an automated diabetic retinopathy detection system using a hybrid deep learning model combined with clinical feature extraction. The overall framework consists of several stages including dataset acquisition, image preprocessing, clinical feature extraction, deep feature extraction, feature fusion, and classification. These stages work together to improve the detection accuracy of diabetic retinopathy from retinal fundus images.

Deep learning models, particularly convolutional neural networks (CNNs), have demonstrated strong performance in medical image analysis because they can automatically learn complex hierarchical features from images. However, many existing methods rely only on deep learning features and ignore clinically meaningful image characteristics. Recent studies highlight

that combining deep learning features with clinical image features can improve the accuracy and reliability of disease detection systems [7][8]. Therefore, the proposed system integrates EfficientNet-B3 and ConvNeXt architectures along with clinical feature extraction to enhance diabetic retinopathy classification performance.

1. DATASET DESCRIPTION

The dataset used in this project is the APTOS 2019 Blindness Detection dataset, which is publicly available on Kaggle. The dataset contains retinal fundus images labeled according to the severity of diabetic retinopathy. Each image has been graded by expert ophthalmologists and categorized into five classes representing different stages of the disease.

These classes include No Diabetic Retinopathy, Mild, Moderate, Severe, and Proliferative diabetic retinopathy. The dataset contains variations in illumination, image quality, and retinal structures because the images were collected from real clinical environments. Such variations make the dataset suitable for training robust deep learning models capable of handling real-world conditions.

Table 1. Dataset Description

Attribute	Description
Dataset Name	APTOS 2019 Blindness Detection
Source	Kaggle
Image Type	Retinal Fundus Images
Classes	5 DR Severity Levels
Training Images	3662
Test Images	1928

2. DATASET ACQUISITION

The retinal images are organized according to their corresponding diabetic retinopathy classes. Each image is labeled with a diagnosis value from 0 to 4, representing increasing severity levels of the disease. To train the deep learning model effectively, the dataset is divided into training, validation, and testing sets.

The training dataset is used to train the model, the validation dataset is used to monitor performance during training, and the testing dataset is used to evaluate the final model. Stratified sampling is used during dataset splitting to ensure balanced representation of all classes.

Table 2. Dataset Acquisition

Class	DR Stage	Train	Validation	Test	Total
0	No DR	1264	271	270	1805
1	Mild	259	56	55	370
2	Moderate	699	150	150	999
3	Severe	135	29	29	193
4	Proliferative	206	44	45	295

3. SYSTEM BLOCK DIAGRAM

The overall workflow of the proposed diabetic retinopathy detection system is shown in Fig. 1. The system consists of several stages including dataset acquisition, image preprocessing, clinical feature extraction, deep feature extraction, feature fusion, and classification. These stages work together to detect and classify different severity levels of diabetic retinopathy from retinal fundus images.

Initially, retinal images are collected from the APTOS 2019 Blindness Detection dataset, which contains labeled fundus images representing five stages of diabetic retinopathy. Since the images are captured under different clinical conditions, preprocessing techniques are applied to improve image quality. Resizing, Gaussian blur, and Ben Graham enhancement are used to highlight retinal structures such as blood vessels and lesions.

After preprocessing, clinical features such as mean intensity, standard deviation, vessel strength, bright pixel ratio, dark pixel ratio, and entropy are extracted. The preprocessed images are then passed through EfficientNet-B3 and ConvNeXt-Tiny networks to extract deep features. These deep features are fused with clinical features and passed through fully connected layers to classify the severity of diabetic retinopathy.

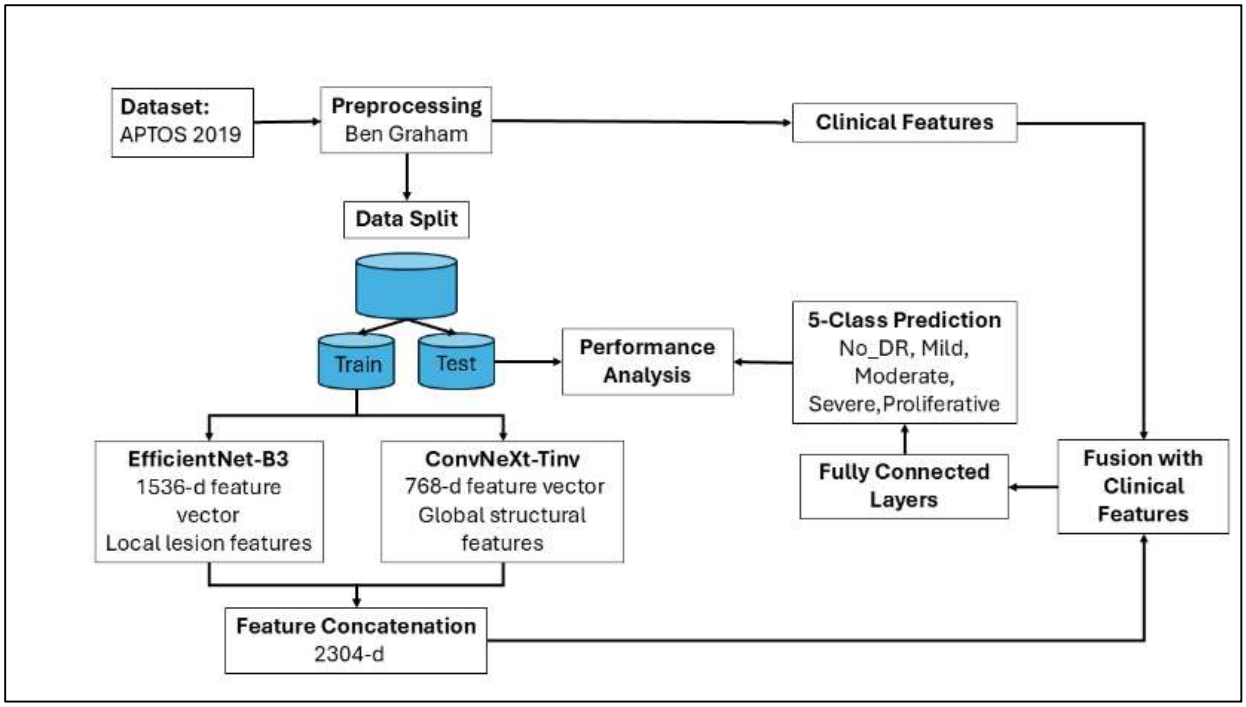

Fig 1. Proposed Model Block Diagram

4. IMAGE PREPOSSING

Image preprocessing is performed to improve the quality of retinal fundus images before feature extraction and classification. Fundus images often contain noise, uneven illumination, and background variations that may affect the performance of deep learning models. Therefore, preprocessing techniques are applied to enhance retinal structures and highlight disease-related features such as blood vessels, microaneurysms, and lesions. In this work, Ben Graham enhancement is applied to improve image contrast and remove background illumination. This method enhances important retinal patterns by subtracting a blurred version of the image from the original image.

The Ben Graham enhancement formula is given by:

$$I_{enhanced} = \alpha I - \beta G(I) + \gamma \tag{1}$$

where

I= original image

$G(I)$= Gaussian blurred image

$$\alpha = 4,\ \beta = -4,\ \gamma = 128$$

To obtain the blurred image, Gaussian blur is applied to the retinal image. Gaussian blur reduces noise by smoothing the image using a Gaussian distribution.

The Gaussian blur formula is:

$$G(I) = I * \mathcal{N}(0, \sigma^2) \tag{2}$$

where

$*$represents convolution,

$\mathcal{N}(0, \sigma^2)$represents the Gaussian kernel with mean 0 and variance σ^2.

These preprocessing techniques enhance retinal features and improve the ability of the deep learning model to detect diabetic retinopathy accurately.

5. CLINICAL FEATURE EXTRACTION

Along with deep learning features, several clinical features are extracted from retinal images to capture statistical characteristics of retinal structures. These features include mean intensity, standard deviation, vessel strength, bright pixel ratio, dark pixel ratio, and image entropy.

Mean intensity and standard deviation represent brightness and intensity variation in the image. Vessel strength is calculated using edge detection techniques to capture retinal blood vessel structures. Bright and dark pixel ratios indicate the presence of lesions or hemorrhages, while entropy measures texture complexity in the retinal image. Combining clinical features with deep learning features improves the interpretability and robustness of the automated diabetic retinopathy detection system [9][11].

Table 3. Extracted Clinical Features

Feature	Description
Mean Intensity	Average brightness of retinal image
Standard Deviation	Intensity variation
Vessel Strength	Edge-based vessel detection
Bright Pixel Ratio	Bright lesion proportion
Dark Pixel Ratio	Dark lesion proportion
Entropy	Texture complexity of image

6. HYBRID MODEL ARCHITECTURE

The proposed model utilizes a hybrid deep learning architecture that integrates two powerful neural network models, EfficientNet-B3 and ConvNeXt-Tiny, to extract deep features from retinal fundus images. Using multiple architectures helps capture different levels of visual information and improves the overall performance of diabetic retinopathy detection. EfficientNet-B3 is used to extract high-level semantic features from retinal images. It is a highly efficient convolutional neural network that balances model depth, width, and resolution to achieve better performance in image classification tasks. The architecture of EfficientNet-B3 used in this study is shown in Fig. 2.

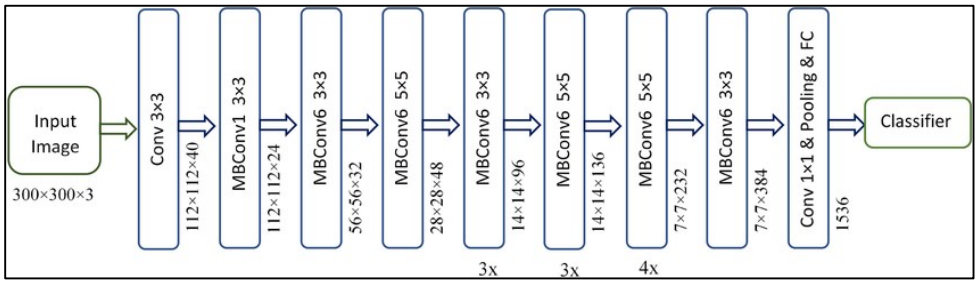

Fig 2. Architecture of EfficientNet-B3

In addition to EfficientNet-B3, the ConvNeXt-Tiny architecture is used to extract complementary feature representations from retinal images. ConvNeXt is a modern convolutional neural network inspired by transformer-based architectures and is designed to improve feature extraction capabilities. It captures additional spatial and structural patterns present in retinal images that may not be fully captured by a single network. The architecture of ConvNeXt-Tiny used in the proposed system is illustrated in Fig. 2

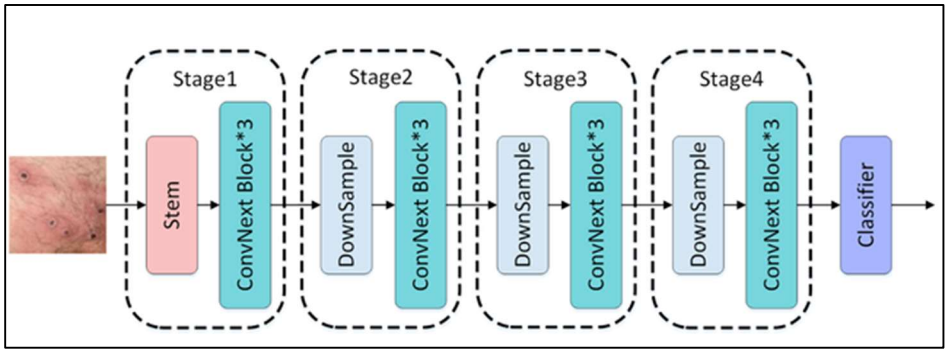

Fig 3. Architecture of ConvNeXt - Tiny

The deep features extracted from both EfficientNet-B3 and ConvNeXt-Tiny are then combined with the clinical features extracted from the retinal images. This process is known as feature fusion and helps create a comprehensive representation of retinal disease patterns. The EfficientNet-B3 model extracts 1536 deep features, while the ConvNeXt-Tiny model extracts 768 deep features. Along with these, 6 clinical features are included, resulting in a total fused feature vector of 2310 features. This combined feature representation is then used as input for the classification layer to predict the severity level of diabetic retinopathy.

RESULTS

The performance of the proposed hybrid deep learning model was evaluated using standard classification metrics including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive evaluation of the model's ability to correctly classify retinal fundus images into different diabetic retinopathy severity levels. The proposed model combines EfficientNet-B3, ConvNeXt-Tiny, and clinical feature extraction, allowing the system to learn both deep visual patterns and clinically relevant image characteristics.

The experimental results show that the proposed hybrid model achieved an overall accuracy of 85.82% for 5-class diabetic retinopathy classification using the APTOS 2019 dataset. The classification report presented in Table 4 shows the detailed performance for each class. The model achieves very high precision and recall values for the No_DR class, indicating that normal retinal images are detected with high reliability. The Moderate class also shows balanced performance with similar precision and recall values. However, the Severe and Proliferative classes show slightly lower performance due to class imbalance and fewer training samples available for these stages of the disease.

Table 4. Classification Report of Proposed Model

Class / Metric	Precision	Recall	F1-Score	Support
No_DR	0.99	0.99	0.99	271
Mild	0.78	0.70	0.74	56
Moderate	0.79	0.79	0.79	150
Severe	0.56	0.62	0.59	29
Proliferative	0.60	0.64	0.62	44

Accuracy	–	–	0.86	550
Macro Avg	0.74	0.75	0.74	550
Weighted Avg	0.86	0.86	0.86	550

The results also include macro average and weighted average metrics, which provide an overall summary of the model performance across all classes. The weighted average values indicate strong overall classification capability of the proposed system.

To further evaluate the effectiveness of the proposed model, a comparison was conducted with several existing diabetic retinopathy detection approaches. The comparison results are shown in Table 5. The proposed hybrid model achieves higher accuracy compared to previously reported methods such as Yi et al., Balaji et al., and Karthik et al. The improvement in performance demonstrates that combining multiple deep learning architectures with clinical feature extraction helps improve classification accuracy.

Table 5. Comparison of Proposed Hybrid Model with Existing Methods

Method	**Class**	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 Score (%)**
Initial Model	Binary	98.55	98.22	98.42	98.57
	5-Class	83.30	82.85	83.30	82.88
Initial Model With Quantum	Binary	98.00	98.00	96.00	98.00
Proposed Model	Binary	98.36	98.37	98.36	98.36
	5-Class	85.82	86.00	85.82	85.87
(S. Balaji et al.,2024) hybrid model	5-Class	82.20	82.10	83.00	82.00

The comparison results show that the proposed model achieves 85.82% accuracy, which is higher than the previously reported methods. The improvement is mainly due to the use of hybrid feature extraction, which captures both deep learning features and clinical image features.

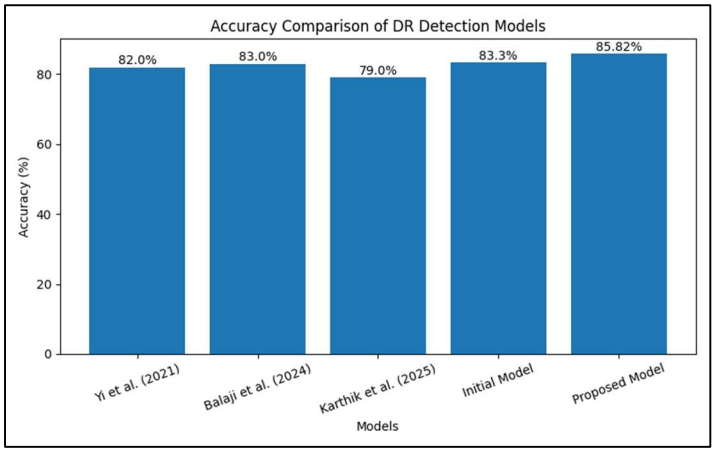

Fig 4. Comparison Graph of Proposed Model

The comparison graph illustrates the accuracy achieved by different diabetic retinopathy detection models. It clearly shows that the proposed hybrid model achieves the highest accuracy among the compared methods, demonstrating the effectiveness of the proposed architecture.

The confusion matrix shown in Fig. 5 provides a detailed visualization of the classification performance of the proposed model across the five diabetic retinopathy classes. Each row of the matrix represents the actual class, while each column represents the predicted class. The values along the diagonal represent correctly classified samples. From the confusion matrix, it can be observed that most of the predictions lie along the diagonal, indicating that the majority of retinal images are correctly classified. The No_DR class shows the highest number of correct predictions due to the larger number of training samples available. Some misclassifications occur between Mild, Moderate, Severe, and Proliferative classes, mainly because these stages share similar retinal features and the dataset contains fewer samples for advanced disease stages.

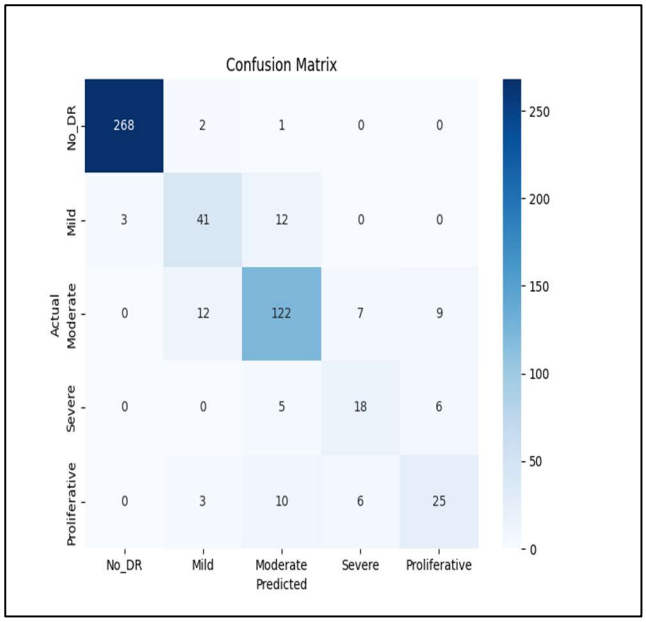

Fig 5. Proposed Model 5-Class Confusion Matrix

The confusion matrix demonstrates that the proposed hybrid deep learning model is capable of effectively distinguishing between different stages of diabetic retinopathy. Despite minor misclassifications between adjacent disease stages, the model shows strong overall classification performance and reliable detection capability.

Statement of Contribution

This interim progress report, and the activities that it has entailed, has been a collaborative effort amongst the group members.

SUJEET SWARAN A S

- Contributed to data collection and dataset preparation
- Performed image preprocessing and enhancement techniques
- Assisted in report documentation

DEENAN M

- Implemented deep learning models (EfficientNet & MobileViT)
- Worked on feature fusion and LVAM attention module
- Conducted training and performance evaluation

MOHAMED FAYAZ S

- Developed clinical feature extraction methods
- Helped in integrating multimodal features into the model
- Prepared presentation materials and results visualization

Declaration

SUJEET SWARAN A S	:
DEENAN M	:
MOHAMED FAYAZ S	:
Project Supervisor Dr. ANILET BALA A	: